

Expression of a single inhibitory Ly49 receptor is sufficient to license NK cells for effector functions

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Abstract

Natural killer (NK) cells recognize target cells through germline-encoded activation and inhibitory receptors enabling effective immunity against viruses and cancer. The Ly49 receptor family in the mouse and killer immunoglobulin-like receptor family in humans play a central role in NK cell immunity through recognition of MHC class I and related molecules. Functionally, these receptor families are involved in licensing and rejection of MHC-I-deficient cells through missing-self. The Ly49 family is highly polymorphic, making it challenging to detail the contributions of individual Ly49 receptors to NK cell function. Herein, we showed mice lacking expression of all Ly49s were unable to reject missing-self target cells *in vivo*, were defective in NK cell licensing, and displayed lower KLRG1 on the surface of NK cells. Expression of Ly49A alone on a H-2D^d background restored missing-self target cell rejection, NK cell licensing, and NK cell KLRG1 expression. Thus, a single inhibitory Ly49 receptor is sufficient to license NK cells and mediate missing-self *in vivo*.

eLife assessment

This **valuable** study presents **compelling** evidence that a single member of the Ly49 gene family (Ly49a) provides sufficient inhibitory signaling to license NK cell activity when its H-2Dd ligand is present. There is also **convincing** evidence of the effect of Ly49a expression on *in vitro* killing and IFN γ production. The use of the authors' system to investigate additional Ly49 receptors, such as Ly49c/i on the H2b background, could provide information on their relative contribution to NK cell licensing. Improvements to the presentation with respect to figure clarity and terminology would allow a better understanding of this complex system by non-experts.

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Introduction

Natural killer (NK) cells are innate lymphoid cells (ILCs) that can mediate effective immunity against viruses and cancer through direct lysis and cytokine production (Huntington et al., 2020; Piersma and Brzić, 2021). NK cells recognize their target cells through integration of signals by germline-encoded activation and inhibitory receptors (Long et al., 2013). These inhibitory receptors include members of the Ly49 family in the mouse and killer immunoglobulin-like receptor (KIR) family in humans and they prevent killing of healthy cells through recognition of MHC class I (MHC-I) (Colonna and Samaridis, 1995; Karlhofer et al., 1992). Host cells may lose surface MHC-I expression in response to virus infection or malignant transformation. As a result, these cells become invisible to CD8⁺ T cells, but simultaneously become targets for NK cells through “missing-self” recognition (Kärre et al., 1986).

The inhibitory Ly49 molecules appear to be responsible for missing-self recognition in mice (Babić et al., 2010; Belanger et al., 2012; Gamache et al., 2019; Parikh et al., 2020; Zhang et al., 2019). However, it has been sometimes challenging to draw definitive conclusions because the Ly49 family is highly polymorphic and differs between mouse strains. Moreover, multiple inhibitory Ly49 receptors within a single host can recognize a given MHC-I molecule while others apparently have no ligands and instead recognize other MHC-I alleles (Schenkel et al., 2013). Yet, the Ly49s for non-host MHC-I alleles are still expressed. For example, in the C57BL/6 background, Ly49C and Ly49I can recognize H-2^b MHC-I molecules, while Ly49A and Ly49G cannot recognize H-2^b molecules and instead they recognize H-2^d alleles. Still these Ly49s are expressed in C57BL/6 mice, so their individual contributions to missing-self rejection are unclear. Ly49A has also been implicated in recognition of the non-classical MHC-I molecule H2-M3, which is upregulated in response to exposure to N-formylated peptides (Andrews et al., 2012; Chiu et al., 1999). Importantly, the specificities of several Ly49s have been clearly established while others remain to be confirmed. For example, the binding of Ly49A to H-2D^d has been confirmed by crystallographic studies (Tormo et al., 1999), and validated by mutational analysis of both Ly49A and H-2D^d (Matsumoto et al., 2001; Wang et al., 2001). By contrast, the MHC-I specificities of other Ly49s have been primarily studied with MHC tetramers containing human β_2 -microglobulin (B2m) which is not recognized by Ly49A (Mitsuki et al., 2004) on cells overexpressing Ly49s (Hanke et al., 1999). Thus, the contributions of individual Ly49 receptors to NK cell effector function are confounded by expression of multiple receptors, some of which may be irrelevant to a given self-MHC haplotype, and multiple Ly49 alleles whose specificities are less well defined.

In addition to effector function in missing-self, when inhibitory Ly49 receptors recognize their cognate MHC-I ligands *in vivo*, they license or educate NK cells for potent effector functions including IFN γ production and degranulation in response to activation receptor stimulation (Elliott et al., 2010; Kim et al., 2005). Like missing-self recognition, inhibitory Ly49s require SHP-1 for NK cell licensing which interacts with the ITIM-motif encoded in the cytosolic tail of inhibitory Ly49s (Bern et al., 2017; Kim et al., 2005; Viant et al., 2014). Moreover, lower expression of SHP-1, particularly within the immunological synapse, is associated with licensed NK cells (Schmied et al., 2023; Wu et al., 2021). Thus, inhibitory Ly49s have a second function that licenses NK cells to self-MHC-I thereby generating appropriate self-tolerant NK cells, but it has not been possible to exclude contributions from other co-expressed Ly49s.

The complex nature of the Ly49 family confounds our fundamental understanding of these receptors, particularly regarding their function *in vivo*. To better understand Ly49 function *in vivo*, several groups made mutant mouse lines with altered expression of the *Ly49* locus. This was pioneered by the Makrigiannis group, which targeted the *Ly49o* promoter in 129 ES cells and resulting mice were subsequently backcrossed to C57BL/6 background (Belanger et al., 2012). These mice were defective in rejection of MHC-I deficient target cells *in vivo* and exhibited reduced

tumor control as well (Tu et al., 2014 [↗](#)). However, there was limited surface expression of NKG2A as well as Ly49s, so these target defects were not solely dependent on absence of Ly49s. Moreover, these mice likely carried 129 alleles of other genes in the NK gene complex (NKC) that are expressed on NK cells, display allelic polymorphisms and are genetically linked to *Ly49*. Following the development of mouse CRISPR engineering, the Dong group deleted the entire 1.4 Mb *Ly49* locus in the C57BL/6 NKC and showed that Ly49-deficient mice were unable to reject MHC-I deficient cells *in vivo* at steady state (Zhang et al., 2019 [↗](#)). Besides loss of Ly49 expression, surface expression of other receptors, including KLRG1, NKG2A, NKG2D and CD94 were also reduced in these mice. Around the same time, we generated a mouse that contains a 66 Kb and 149 Kb deletion in the C57BL/6 *Ly49* locus, resulting in loss of 4 Ly49 molecules including Ly49A and Ly49G (Parikh et al., 2020 [↗](#)). The resulting Δ Ly49-1 mice were also deficient in H2D^d-restricted control of murine cytomegalovirus (MCMV), which was rescued by knock-in of *Ly49a* into the *Ncr1* locus. Thus, available genetic evidence suggest inhibitory Ly49 receptors are essential for licensing and missing-self recognition.

The current data, however, do not take into account that individual Ly49s are stochastically expressed on NK cells, and multiple receptors are simultaneously expressed on individual NK cells, resulting in a diverse Ly49 repertoire of potential specificities on overlapping subsets of NK cells (Dorfman and Raulet, 1998 [↗](#); Kubota et al., 1999 [↗](#); Smith et al., 2000 [↗](#)). As a result, not all NK cells express a specific Ly49 receptor and most NK cells express multiple Ly49 molecules, making it difficult to study the biology of a specific Ly49 receptor without genetic approaches. However, *Ly49* genes are highly related and clustered together, resulting in a high concentration of repetitive elements (Makrigiannis et al., 2005 [↗](#)), complicating the capacity to target individual *Ly49* genes for definitive analysis. Moreover, in Δ Ly49-1 mice which had an intact *Ly49d* coding sequence, the percentage of NK cells expressing Ly49D was markedly reduced, even though Ly49D was otherwise expressed at normal levels, suggesting a regulatory locus control region within the deleted fragments (Parikh et al., 2020 [↗](#)). Such data raise the possibility that the large genetic deletions of the *Ly49* locus may affect other NK cell receptors in the NKC that contribute to NK cell function. Thus, expression of individual Ly49 receptors without confounding effects of other Ly49s and potentially other NKC genes is needed to validate the conclusions from study of mice lacking Ly49 expression.

Here we studied the role of an individual Ly49 receptor in NK cell function. To this end, we deleted all NK cell Ly49 genes using CRISPR/Cas9 and confirmed the role of the *Ly49* family in missing self and licensing. Subsequently, we expressed Ly49A in isolation under control of the *Ncr1* locus in Ly49-deficient mice on a H-2D^d background to show that a single inhibitory Ly49 receptor expressed by all NK cells is sufficient for licensing and mediating missing-self *in vivo*.

Results

NK cells from CRISPR-generated mice lacking all NK cell-related Ly49 molecules display reduced KLRG1 expression

To investigate the role of individual Ly49 molecules, we generated a mouse that lacked all expressed Ly49 receptors. We targeted the remaining *Ly49* region in our previously published Δ Ly49-1 mouse (Parikh et al., 2020 [↗](#)) with guide RNAs targeting *Ly49i* and *Ly49q*. The resulting mouse contained a fusion between *Ly49i* and *Ly49q* with a deletion of the start codon and insertion of a fusion sequence that did contain a potential start site for a putative 5 amino acid polypeptide (Figure 1A [↗](#)). We confirmed that the 3' deletion reported in the Δ Ly49-1 mouse was unaffected, resulting in a frameshift and a premature stop codon after 9 amino acids in the fused *Ly49a/g* gene. Thus, genetic and sequencing analysis revealed all Ly49 genes were disrupted and we termed this mouse line Ly49KO (Figure 1A [↗](#)).

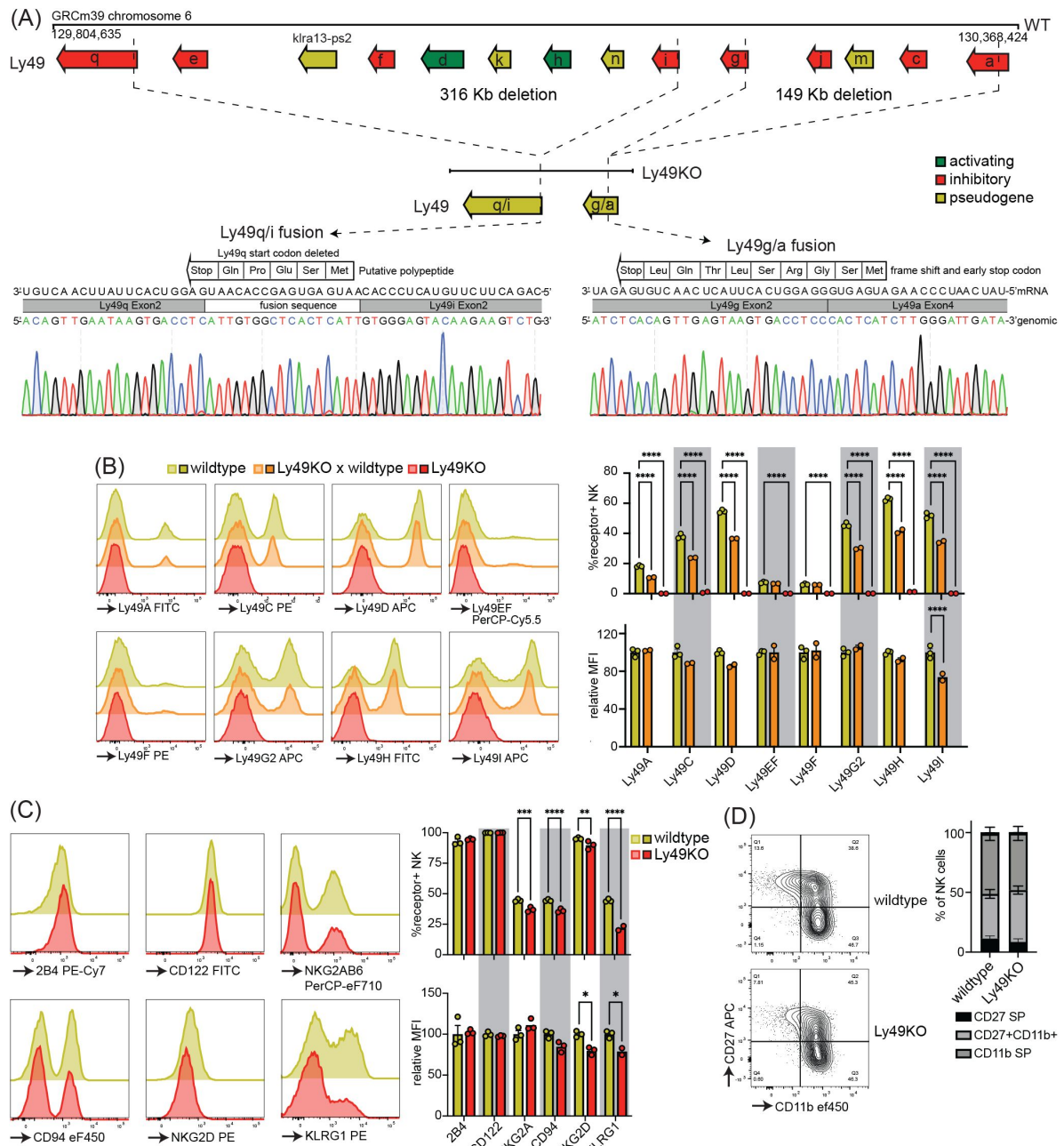


Figure 1.

Mice generated to lack all NK-related Ly49 molecules using CRISPR have NK cells that display a less mature phenotype and impaired KLRG1 expression.

(A) Genetic map of the Ly49 locus of wildtype C57BL/6 and Ly49KO mice and Sanger sequencing of the fusion sequences in the Ly49KO mice. (B) Ly49 receptor expression on splenic NK cells of the indicated mice. (C) Surface receptor expression on splenic NK cells from indicated mice. (D) Expression of the maturation markers CD27 and CD11b on splenic NK cells from indicated mice. MFI, median fluorescent intensity. Error bars indicate SEM; ns, not significant; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

Flow cytometry confirmed loss of cell surface expression of Ly49 molecules in homozygous Ly49KO mice (**Figure 1B**). NKG2A, CD94, and NKG2D molecules that are encoded by the NKG2 locus, located next to the Ly49 locus, were still expressed, albeit at marginally lower frequencies (**Figure 1C**) (Yokoyama and Plougastel, 2003). Unrelated molecules 2B4 and CD122 were unaffected. In heterozygous Ly49KO mice, the percentages of NK cells expressing Ly49A, Ly49C, Ly49D, Ly49G2, Ly49H, and Ly49I were reduced by 33–41%. The median fluorescent intensity (MFI) for Ly49I was reduced by 26% in Ly49I⁺ NK cells in heterozygous Ly49KO mice, while the other Ly49s did not display significant differences in MFI. Consistent with apparent dependence of normal KLRG1 expression on MHC-I expression (Corral et al., 2000) and previous reports (Zhang et al., 2019), Ly49KO NK cells showed a 51% reduction in KLRG1 expression (**Figure 1C**). NK cells in Ly49KO mice displayed similar maturation to wildtype NK cells, based on expression of the surface markers CD27 and CD11b (**Figure 1D**). Thus, Ly49KO mice specifically lack all Ly49 molecules and display moderate alterations in select surface molecules while showing otherwise normal numbers of apparently mature NK cells.

Ly49-deficient NK cells are defective in licensing and rejection of MHC-I deficient target cells

Inhibitory Ly49-positive NK cells can be licensed through recognition of cognate MHC-I molecules, resulting in a phenotype of increased IFN γ production following plate-bound anti-NK1.1 stimulation (Kim et al., 2005). Stimulation of Ly49KO NK cells anti-NK1.1 resulted in a 73% reduction in IFN γ production as compared to wildtype NK cells, similar to unlicensed NK cells from MHC-I-deficient H-2K^b x H-2D^b deficient (KODO) mice (**Figure 2A**). Both Ly49KO and KODO NK cells produced high amounts of IFN γ in response to phorbol 12-myristate 13-acetate (PMA) plus ionomycin (**Figure 2B**), indicating that their IFN γ production machinery is intact. Thus, these results confirm that Ly49 molecules are required for the NK cell licensed phenotype.

To investigate the capability of Ly49KO NK cells to reject MHC-I deficient target cells, we challenged anti-NK1.1 NK cell-depleted, wildtype, and Ly49KO mice with a mixture of wildtype, B2m deficient x KODO (MHC-I KO), H-2D^b KO, and H-2K^b KO splenocytes that were differentially labeled with CellTrace Far Red (CTFR) and CellTrace Violet (CTV) (**Figure 2C**). While wildtype mice efficiently rejected MHC-I KO and H-2K^b KO target cells, only 13% of H-2D^b KO target cells were rejected by wildtype mice, indicating that missing-self is dominated by H-2K^b in the C57BL/6 background. None of the target cell populations were rejected in the Ly49KO mice, comparable to wild type controls depleted of NK cells. Thus, Ly49 molecules mediate NK cell-dependent MHC-I deficient target cell killing *in vivo* under steady-state conditions and rejection of MHC-I deficient target cells is predominantly controlled by H-2K^b in the H-2^b background.

Expression of Ly49A in Ly49-deficient H-2D^d transgenic mice rescues KLRG1 expression

To investigate the potential of a single inhibitory Ly49 receptor on mediating NK cell licensing and missing-self rejection, the Ly49KO mice were backcrossed to H-2D^d transgenic KODO (D8-KODO) Ly49A KI mice that express *Klra1* cDNA encoding the inhibitory Ly49A receptor in the *Ncr1* locus and its cognate ligand H-2D^d but not any other classical MHC-I molecules (Parikh et al., 2020). Ly49A expression in the resulting Ly49KO/Ly49A KI D8-KODO mice closely follows NKp46 expression because NKp46⁺ NK1.1⁺ NK cells in the bone marrow of these mice do not express Ly49A, while virtually all the NKp46⁺ NK1.1⁺ NK cells in the bone marrow and spleen express Ly49A (**Figure 3A**). Ly49KO/Ly49A KI D8-KODO NK cells expressed robust levels of Ly49A, albeit at lower MFI as compared to Ly49A expression on D8-KODO NK cells, consistent with prior observations with wild type Ly49A and H2D^d (Held et al., 1996; Karlhofer et al., 1994). NK cells were able to fully mature in Ly49KO D8-KODO and Ly49KO/Ly49A KI D8-KODO mice as we observed similar percentage of mature CD27⁺ CD11b⁺ NK cells in spleen, bone marrow, and liver (**Figure 3B**). While there was a modest significant increase in immature CD27⁺ CD11b⁺ NK cells

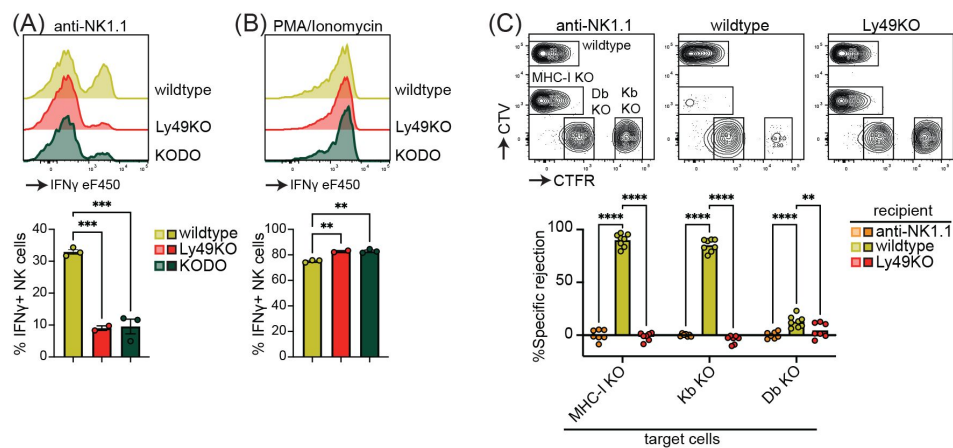


Figure 2.

NK cell licensing and rejection of MHC-I deficient target cells is defective in Ly49KO mice.

Splenocytes from the indicated mice were stimulated with plate bound anti-NK1.1 **(A)** or PMA/ionomycin **(B)** and IFN γ production by NKG2A-NK cells and analyzed by flow cytometry. **(C)** *In vivo* cytotoxicity assay against H-2K^b, H-2D^b and full MHC-I deficient targets. Splenocytes from WT, H-2K^b, H-2D^b and MHC-I deficient mice were differentially labeled with CFSE, CTV, and CTFR as indicated. Mixture of labeled target cells were injected i.v. into wildtype, Ly49KO, and anti-NK1.1-depleted mice. Target cells were analyzed in spleens by flow cytometry 2 days after challenge. KODO, H-2K^b x H-2D^b knock out; MHC-I KO, KODO x B2m knockout; MFI, median fluorescent intensity. Error bars indicate SEM; ns, not significant; * p < 0.05, ** p < 0.01, *** p < 0.001, and **** p < 0.0001.

in the bone marrow of Ly49KO D8-KODO and Ly49KO/Ly49A KI D8-KODO mice, no differences were observed in NK cell maturation in spleen and liver. The decrease in the frequency of KLRG1⁺ NK cells observed in the Ly49KO NK cells on the H-2^b background was recapitulated in the D8-KODO background (**Figure 3C**). Intriguingly, expression of Ly49A in Ly49KO/Ly49A KI D8-KODO mice rescued KLRG1 expression and resulted in similar levels of KLRG1 as D8-KODO NK cells. Taken together, Ly49A engineered to be encoded within the Nkp46 locus was efficiently expressed as an isolated Ly49 receptor and supported KLRG1 expression.

NK cells expressing Ly49A in isolation are fully licensed and capable of rejecting MHC-I deficient target cells

Next, we investigated the potential of Ly49A expression alone to mediate NK cell licensing and rejection of MHC-I deficient target cells. In D8-KODO mice, Ly49A⁺ NK cells displayed increased levels of IFN γ production and degranulation measured by CD107 in response to plate bound anti-NK1.1 stimulation as compared to all NK cells including unlicensed cells (**Figure 4A**). Similar to Ly49KO NK cells on the H-2^b background, Ly49KO NK cells on the D8-KODO background showed a 72% decrease in IFN γ production, but also showed a 59% decrease in degranulation as compared to Ly49A⁺ NK cells in D8-KODO mice. This impaired IFN γ production and degranulation was reversed in Ly49A KI NK cells on the Ly49KO D8-KODO background, that showed similar IFN γ production and degranulation to Ly49A⁺ NK cells on the D8-KODO background, indicating that Ly49A KI NK cells are licensed. Importantly, there were no differences among all NK cell populations in IFN γ production and degranulation in response to PMA/Ionomycin (**Figure 4B**), indicating that the IFN γ production and degranulation machinery are not affected in any of the mouse strains. Thus, the Ly49KO/Ly49A KI D8-KODO NK cells displayed a fully licensed phenotype comparable to licensed Ly49A⁺ NK cells in D8-KODO mice.

Finally, we interrogated whether a single inhibitory Ly49 molecule would be sufficient to mediate missing-self rejection of MHC-I deficient cells. To this end D8-KODO, Ly49KO D8-KODO, and Ly49KO/Ly49A KI D8-KODO mice were challenged with a mixture of D8-KODO and KODO (MHC-I deficient) splenocytes that were differentially labelled with CTV. D8-KODO mice efficiently rejected KODO target splenocytes, while Ly49KO D8-KODO mice were unable to reject these cells. However, this defect was completely restored in the Ly49KO/Ly49A KI D8-KODO mice, demonstrating that a single inhibitory Ly49 receptor is sufficient to mediate missing-self rejection of cells lacking its MHC class I ligand.

Discussion

Several mice with genetic modifications in the Ly49 complex have been developed to study the role of Ly49 receptors but they have limitations (Belanger et al., 2012; Bern et al., 2017; Gamache et al., 2019; Parikh et al., 2020; Zhang et al., 2019). A complicating factor in these studies is that multiple Ly49s, often with incompletely understood specificities, may be involved in target cell recognition. Here, we studied a mouse where a single Ly49 was under control of the *Ncr1* locus which is expressed on all NK cells on the background of a complete Ly49 KO. Consistent with previous reports (Belanger et al., 2012; Parikh et al., 2020; Zhang et al., 2019), Ly49-deficient NK cells were deficient in licensing and missing-self rejection, both on a H-2^b background and in the presence of a single classical MHC-I allele, H-2D^d. Moreover, expression of the inhibitory Ly49A in isolation did not alter NK cell numbers or maturation, indicating that the Ly49s do not affect these parameters of NK cells. Yet the NK cells were fully licensed in terms of their functional phenotype, including capacity to be activated by an activation receptor *in vitro* and efficient rejection of MHC-I deficient target cells *in vivo*. Thus, a single Ly49 receptor confers the licensed phenotype *in vitro* and *in vivo*.

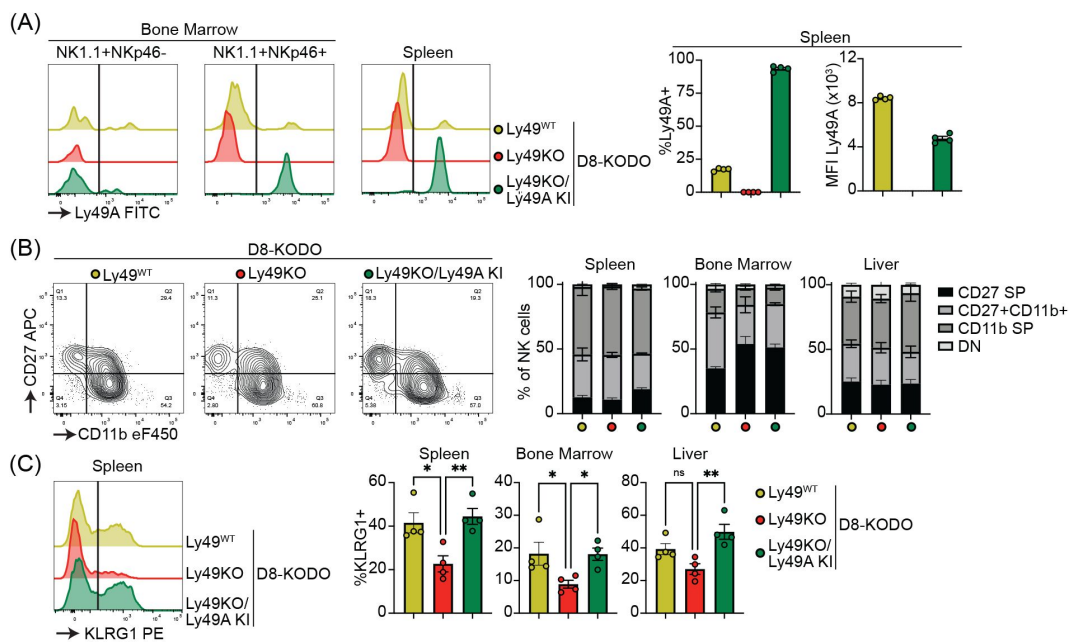


Figure 3.

Ly49A is efficiently expressed in Ncr1-Ly49A knockin mice and rescues KLRG1 expression in NK cells.

Flow cytometric analysis of NK cells in D8-KODO, Ly49KO D8-KODO, and Ly49KO/Ly49A KI D8-KODO mice **(A)** Ly49A expression in NKp46⁺ and NKp46⁻ NK1.1⁺ NK cells in bone marrow and spleen of the indicated mice. **(B)** Expression of the maturation markers CD27 and CD11b on NK cells in spleen, bone marrow, and liver of indicated mice. **(C)** KLRG1 expression by NK cells in spleen, bone marrow, and liver of indicated mice. MFI, median fluorescent intensity. Error bars indicate SEM; ns, not significant; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

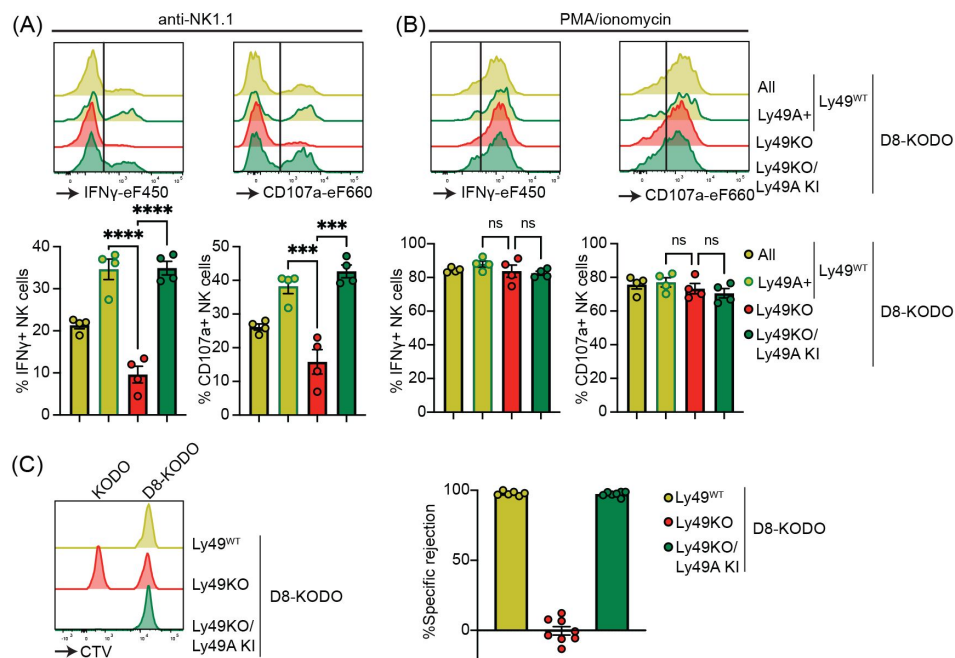


Figure 4.

Expression of Ly49A in isolation is sufficient for NK cell licensing and missing-self rejection.

Splenocytes from the indicated mice were stimulated with plate bound anti-NK1.1 (A) or PMA/ionomycin (B). IFN γ production and degranulation (CD107a) by NK2A-NK cells were analyzed by flow cytometry. (C) Splenocytes from D8-KODO and KODO mice were differentially labeled with CFSE and CTV as indicated. A mixture of labeled target cells was injected i.v. into D8-KODO, Ly49KO D8-KODO, and Ly49KO/Ly49A KI D8-KODO mice. Specific rejection of target cells was analyzed in spleens by flow cytometry 2 days after challenge. Error bars indicate SEM; ns, not significant; *** $p < 0.001$, and **** $p < 0.0001$.

We observed that rejection of H-2K^b-deficient targets was more potent than H-2D^b-deficient target splenocytes, comparable to previous observations (Johansson et al., 2005 [\[1\]](#)). These data indicate that H-2K^b is more efficiently recognized by Ly49s as compared to H-2D^b. On the other hand, early studies using Ly49 transfectants binding to Con A blasts found that Ly49C and Ly49I can bind to H-2D^b-deficient but not H-2K^b-deficient cells (Hanke et al., 1999 [\[2\]](#)), but these studies have the caveat of testing binding to cells overexpressing Ly49s. Our studies indicate that the efficiency of Ly49-dependent missing-self rejection depends on characteristics of specific MHC-I alleles recognized by cognate Ly49 receptors *in vivo*, raising a caution to solely depending on Ly49 specificities based on *in vitro* studies alone.

KLRG1 is an inhibitory receptor that recognizes E-, N-, and R-cadherins to inhibit NK cell cytotoxicity (Ito et al., 2006 [\[3\]](#)). KLRG1 is expressed on a subset of mature NK cells and can be upregulated in response to proliferation in a host with lymphopenia (Huntington et al., 2007 [\[4\]](#)). Consistent with previously published results (Zhang et al., 2019 [\[5\]](#)), we observed decreased KLRG1 expression in Ly49-deficient NK cells. The *Ly49* gene family as well as *Klrg1* is located within the NKC (Yokoyama and Plougastel, 2003 [\[6\]](#)), thus an effect of regulatory elements deleted in Ly49KO mice cannot be excluded. This is further emphasized by studies of our ΔLy49-1 mouse which expresses Ly49D on fewer NK cells (Parikh et al., 2020 [\[7\]](#)). The *Ly49d* gene appears intact and Ly49D⁺ NK cells expressed Ly49D at normal levels, suggesting the absence of a regulatory element in the deleted regions. Here, we showed expression of only Ly49A in Ly49KO mice on a H-2D^d background restored KLRG1 expression in NK cells from different tissues, indicating that inhibitory Ly49 receptors rather than regulatory elements influence KLRG1 expression. Moreover, NK cell KLRG1 expression is modulated by MHC-I molecules (Corral et al., 2000 [\[8\]](#)). Therefore, KLRG1 expression may be modulated as a consequence of NK cell licensing through inhibitory Ly49 receptors.

The Ly49 family is stochastically expressed on NK cells, resulting in a NK cell repertoire with different combinations of Ly49 receptors on individual NK cells. Not all Ly49 alleles are equally expressed which has been suggested to be dependent on allelic exclusion (Held et al., 1995 [\[9\]](#)). Epigenetic control including DNA-methylation, histone modification, and regulatory elements have been linked to Ly49 expression (Kissiov et al., 2022 [\[10\]](#); McCullen et al., 2016 [\[11\]](#); Rouhi et al., 2006 [\[12\]](#); Saleh et al., 2004 [\[13\]](#)). We observed that in Ly49KO heterozygous mice the percentage of Ly49⁺ NK cells was reduced for each Ly49 molecule, yet the expression level measured by MFI was not affected except for Ly49I, indicating that loss of one allele does not affect Ly49 surface levels. Nonetheless, alternate mechanisms may control Ly49 expression as was observed for Ly49I. Expression of *Ly49a* within the *Ncr1* locus resulted in ubiquitous Ly49A expression in NK cells, albeit at lower levels compared to Ly49A⁺ D8-KODO NK cells. Despite the lower expression levels of Ly49A, Ly49A KI NK cells were fully licensed and efficiently eliminated MHC-I-deficient target cells, suggesting that minor alterations in expression levels of various Ly49s on individual NK cells may not affect NK cell functions. In conclusion, these data show that expression of a single inhibitory Ly49 receptor is necessary and sufficient to license NK cells for missing self-rejection under steady state conditions *in vivo*.

Materials and Methods

Animals

C57BL/6 (stock # 556) mice were purchased from Charles Rivers laboratories, B2m-deficient (stock # 2087) were purchased from Jackson laboratories, H-2K^b x H-2D^b double-deficient (stock # 4215; KODO) mice were purchased from Taconic Farms. D8 is a H-2D^d transgenic mouse that has been previously described (Bieberich et al., 1986 [\[14\]](#)) and was provided by D. Marguiles, National Institute of Allergy and Infectious Diseases, Bethesda, MD. D8-KODO mice have been previously generated by crossing D8 transgenic mice to a KODO background (Choi et al., 2011 [\[15\]](#)). ΔLy49-1 and

Ly49A KI mice were previously generated in our laboratory (Parikh et al., 2020 [DOI](#)). All mice were maintained within the Washington University animal facility in accordance with institutional ethical guidelines under protocol number 21-0090. All experiments utilized sex- and age-matched mice.

Generation of Ly49KO mice

The remaining Ly49 receptors in Δ Ly49-1 mice were targeted using CRISPR/Cas9 as previously described (Parikh et al., 2015a [DOI](#)). Briefly, the Ly49 locus was targeted with gRNAs directed against Ly49q (5'-ACCCATGATGAGTGAGCAGG-3') and Ly49i (5'-TGAGACTTCATAAGTCTTCAAGG-3'), with the PAM sequence underlined. For the mRNA microinjections, 20ng of each guide and 100ng of Cas9 mRNA were used. Deletions in the Ly49 locus were screened using the primers 5'-GCCCATCTGGCTTCCTTTCT-3' (Ly49q-Rv), 5'-CAAGCCCCGATGAGATGGAT-3' (Ly49i-Rv), and GGATCAGTCCATGTCAGGGTT (Ly49i-Fw) yielding a 409 bp wildtype and a 552 bp mutant band and confirmed using Southern blot analysis (data not shown). To minimize off-target CRISPR/Cas9 effects, candidate founder mice were backcrossed to C57BL/6 mice for 2 generations then crossed to derive homozygous Ly49KO mice. Deletions were verified by Sanger sequencing (Azenta Life Sciences) in homozygous Ly49KO mice using the PCR primers Ly49q-Rv and Ly49i-Fw for the Ly49q/i fusion sequence and the primers AACCAAGCCCCAATGAGATC (Ly49g-Rv) and TGGGTCAGTCCATGTCAGTG (Ly49a-Fw) for the Ly49a/g fusion sequence resulting in 552bp and 409bp products, respectively.

Flow cytometry

Fluorescent-labeled antibodies Ly49D (clone 4D11), Ly49EF (CM4), Ly49F (HBF-719), Ly49G2 (eBio4D11), Ly49H (3D10), Ly49I (YLI-90), 2B4 (eBio244F4), CD122 (TM-b1), NKG2AB6 (16a11), NKG2ACE (20D5), CD94 (18d3), NKG2D (CX5), CD27 (LG.7F9), CD11b (M1/70), IFN γ (XMG1.2), CD107a (eBio1D4B), NKp46 (29A1.4), CD3(145-2C11), CD4 (RM4-5), CD8 (53-6.7), TCRB (H597), and CD19 (eBio1D3) were purchased from Thermo Fisher Scientific; Ly49A (YE1/48.10.6), KLRG1 (2F1), and NK1.1 (PK136), were purchased from Biolegend; Ly49C (4LO311) was purchased from Leinco Technologies. Cells were stained with fixable viability dye eF506 (Thermo Fisher Scientific), continued by staining of cell surface molecules in 2.4G2 hybridoma supernatant to block Fc receptors. For intracellular staining, cells were fixed and stained intracellularly using the BD Cytofix/Cytoperm Fixation/Permeabilization Kit (BD Bioscience) according to manufacturer's instructions. Samples were acquired using FACSCanto (BD Biosciences) and analyzed using FlowJo software (BD Biosciences). NK cells were defined as singlet Viability-NK1.1+NKp46+CD3-CD19- or Viability-NK1.1+NKp46+CD4-CD8-TCRB-CD19-.

In vitro stimulation assays

Stimulation of splenic NK cells was performed as previously described (Parikh et al., 2020 [DOI](#); Piersma et al., 2019 [DOI](#)). Briefly, 1 - 4 μ g/ml anti-NK1.1 (clone PK136, Leinco Technologies) in PBS was coated in 24-well plates for 90 min at 37°C. Plates were washed with PBS and 5 x 10⁶ splenocytes were added per well. In parallel, splenocytes were stimulated with 200 ng/ml Phorbol myristate acetate (PMA; Sigma-Aldrich) and 400 ng/ml Ionomycin (Sigma-Aldrich). After 30 min incubation at 37°C, Monensin (Thermo Fisher Scientific) and fluorescently labelled anti-CD107a antibody were added, cultures were incubated for an additional 7 hours at 37°C and subsequently analyzed by flow cytometry.

In vivo killing assays

In vivo killing assays were performed as previously described (Parikh et al., 2015b [DOI](#)). Briefly, target splenocytes were isolated from C57BL/6, MHC-I deficient (TKO), H-2K^b-deficient, H-2D^b-deficient, KODO and D8-KODO mice. Indicated target splenocytes were differentially labelled with CFSE, CellTrace violet, and/or CellTrace far red (Thermo Fisher Scientific). Target cells were mixed at equal ratios for each target and 2 x 10⁶ splenocytes per target were injected i.v. into naïve hosts.

Where indicated NK cells were depleted with 100µg anti-NK1.1 (Leinco technologies) 2 days before target cell injection. Two days after challenge splenocytes were harvested and analyzed by flow cytometry. Target cell rejection was calculated using the formula $[(1 - (\text{Ratio(KO target/wildtype target)}_{\text{sample}} / \text{Ratio(KO target/wildtype target)}_{\text{control}})) \times 100]$.

Statistics

Statistical analysis was performed with Prism (GraphPad software) using unpaired *t*-tests and two-way ANOVA with corrections for multiple testing. Error bars in figures represent the SEM. Statistical significance was indicated as follows: ****, $p < 0.0001$; ***, $p < 0.001$; **, $p < 0.01$; *, $p < 0.05$; ns, not significant.

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Editors

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Reviewer #1 (Public review):

Summary:

The article by Piersma *et al.* aims to reduce the complex process of NK cell licensing to the action of a single inhibitory receptor for MHC class I. This is achieved using a mouse strain

lacking all of the Ly49 receptors expressed by NK cells and inserting the Ly49a gene into the Ncr1 locus, leading to expression on the majority of NK cells.

Strengths:

The mouse model used represents a precise deletion of all NK-expressed genes within the Ly49 cluster. The re-introduction of the Ly49a gene into the Ncr1 locus allows expression by most NK cells. Convincing effects of Ly49a expression on in vitro activation and in vivo killing assay are shown.

Weaknesses:

The choice of Ly49a provides a clear picture of H-2Dd recognition by this Ly49. It would be valuable to perform additional studies investigating Ly49c and Ly49i receptors for H-2b. This is of interest because there are reports indicating that Ly49c may not be a functional receptor in B6 mice due to strong cis interactions.

This work generates an excellent mouse model for the study of NK cell licensing by inhibitory Ly49s that will be useful for the community. It provides a platform whereby the functional activity of a single Ly49 can be assessed.

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Reviewer #2 (Public review):

Piersma et al. continue to work on deciphering the role and function of Ly49 NK cell receptors. This manuscript shows that a single inhibitory Ly49 receptor is sufficient to license NK cells and eliminate MHC-I-deficient target cells in mice. In short, they refined the mouse model Δ Ly49-1 (Parikh et al., 2020) into the Ly49KO model in which all Ly49 genes are disrupted. Using this model, they confirmed that NK cells from Ly49KO mice cannot be licensed, produce lower levels of IFN- γ , and cannot reject MHC-I-deficient cells. To study the effect of a single Ly49 receptor in the function of NK cells, the authors backcrossed Ly49KO mice to H-2Dd transgenic KODO (D8-KODO) Ly49A knock-in mice in which a single inhibitory Ly49A receptor that recognizes H-2Dd ligands is expressed. By doing so, they demonstrate that a single inhibitory Ly49 receptor expressed by all NK cells is sufficient for licensing and missing-self killing.

While the results of the study are largely consistent with the conclusions, it is important to address some discrepancies. For instance, in the title of Figure 1, the authors state that NK cells in Ly49KO mice compared to WT mice have a less mature phenotype, which is not consistent with the corresponding text in the Results section (lines 170-171) that states there is no difference in maturation. These differences are not evident in Figure 1, panel D. It is crucial to acknowledge these inconsistencies to ensure a comprehensive understanding of the research findings.

In the legend of Figure 2, the text related to panel C indicates the use of dyes to label the splenocytes, and CFSE, CTV, and CTFR were mentioned. However, only CTV and CTFR are shown on the plots and mentioned in the corresponding text in the Results section. Similarly, in the legend of Figure 4, which is related to panel C, the authors write that splenocytes were differentially labeled with CFSE and CTV as indicated; however, in Figure 4, C and the Results section text, there is no mention of CFSE.

The authors should clarify why they assume that KLRG1 expression is influenced by the expression of inhibitory Ly49 receptors and not by manipulations on chromosome 6, where the genes for both KLRG1 and Ly49 receptors are located. However, a better explanation for the possible influence of other inhibitory NK cell receptors still needs to be included. In the

study by Zhang et al. (doi: 10.1038/s41467-019-13032-5 the authors showed the synergized regulation of NK cell education by the NKG2A receptor and the specific Ly49 family members. Although in this study, Piersma and colleagues show the control of MHC-I deficient cells by Ly49A⁺ NKG2A-NK cells in Figure 4., this receptor is not mentioned in the Results or in the Discussion section, so its role in this story needs to be clarified. Therefore, the reader would benefit from more information regarding NKG2A receptor and NKG2A⁺/⁻ populations in their results.

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Reviewer #3 (Public review):

Summary:

In this study, Piersma et al. successfully generated a mouse model with all Ly49 genes knocked out, resulting in the complete absence of Ly49 receptor expression on the cell surface. The absence of Ly49 expression led to the loss of NK cell education/licensing and consequently, a failure in responsiveness against missing-self target cells. The experimental work and findings are partially overlapping with the previous work by Zhang et al. (2019), who also performed knockout of the entire Ly49 locus in mice and demonstrated that loss of NK responsiveness was due to the removal of inhibitory, and not activating Ly49 genes. The authors demonstrate the restoration of NK cell licensing by knocking in a single Ly49 gene, Ly49A, in a mouse expressing the H-2Dd ligand for this receptor, which is a novel and important finding.

Strengths:

The authors established a novel mouse model enabling them to have a clean and thorough study on the function of Ly49 on NK cell licensing. Also, by knocking in a single Ly49, they were able to investigate the function of a given Ly49 receptor excluding the "contamination" of co-expression of any other Ly49 genes. Their idea and method were novel though the mouse model was somehow genetically similar to a previous study. The experiment design and data interpretation were logically clear and the evidence was solid.

Weaknesses:

The paper is very poorly written and confusing. The authors should be more accurate in the usage of terminology, provide more details on experimental procedures, and revise much of the text to improve clarity and coherence. A thorough revision aiming to clarify the paper would be helpful.

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